Estimation of HbA1c Enzymatic Method (Direct Method)

1. Aim

To estimate **HbA1c (%)** in whole blood using a **direct enzymatic method** based on protease digestion and oxidation of glycated valines.

2. Principle

1. **Whole blood is lysed** → haemoglobin is released.
2. **Protease digests haemoglobin** → this frees the **glycated N-terminal valine**, which is the specific part of HbA1c.
3. The released glycated valine is acted upon by a special enzyme called: **Fructosyl-Valine Oxidase (FVO)**
4. FVO **oxidizes the glycated valine**, producing **hydrogen peroxide (H_2O_2)** as a by-product.
5. The amount of **H_2O_2 formed is proportional to the amount of HbA1c** present.
6. H_2O_2 then reacts with a **chromogenic system** to form a coloured compound.
7. The colour intensity is measured in a colorimeter/automated analyser.

3. Clinical Importance

- HbA1c increases when blood glucose stays high over time.
- It reflects **long-term glycemic control**, unlike random blood glucose.
- Chronic hyperglycemia (e.g., in uncontrolled diabetes) produces **higher HbA1c levels**.
- Therefore, HbA1c is used for:
    - Diagnosing diabetes
    - Monitoring diabetic patients
    - Assessing risk of complications

4. Reagents Mentioned in the Source

- **Hemolysis reagent**
- **Protease enzyme** (for hemoglobin digestion)
- **FVO (Fructosyl-Valine Oxidase)**
- **Chromogen + Peroxidase system**
- **Calibrators** (to convert absorbance → % HbA1c)

5. Procedure

1. Take whole blood sample (EDTA).
2. Add hemolysis reagent → mix gently.
3. Add protease reagent → incubate to allow hemoglobin digestion.
4. Add FVO enzyme reagent → oxidation occurs → H_2O_2 forms.
5. Add chromogen/peroxidase reagents → color forms.
6. Measure absorbance at the analyzer's wavelength.
7. Compare absorbance to calibration curve → **report % HbA1c**.

6. Reference Ranges

- **Normal:** around < 6%
- **Increased HbA1c** = poor glycemic control / chronic hyperglycemia

7. Advantages

- **Direct % HbA1c measurement** no total Hb step
- Works well with automated chemistry analyzers
- Accurate, standardized, easy to perform
- Minimal interference after hemolysis and digestion steps
- Clean workflow suitable for routine labs

8. Precautions

- Treat all blood samples as biohazardous.
- Disinfect spills/surfaces (document mentions sodium hypochlorite / autoclaving).
- Do not freeze reagents.
- Do not use reagents after expiry.
- Use reagents **only** for this assay.
- Handle waste, samples, and labware according to biosafety guidelines.